

SRUL: Spatial Spherical Latents with Riemannian Flow Matching

Near-spherical latents make Gaussian sources and straight Euclidean paths geometrically mismatched

Why Euclidean latent dynamics are mismatched

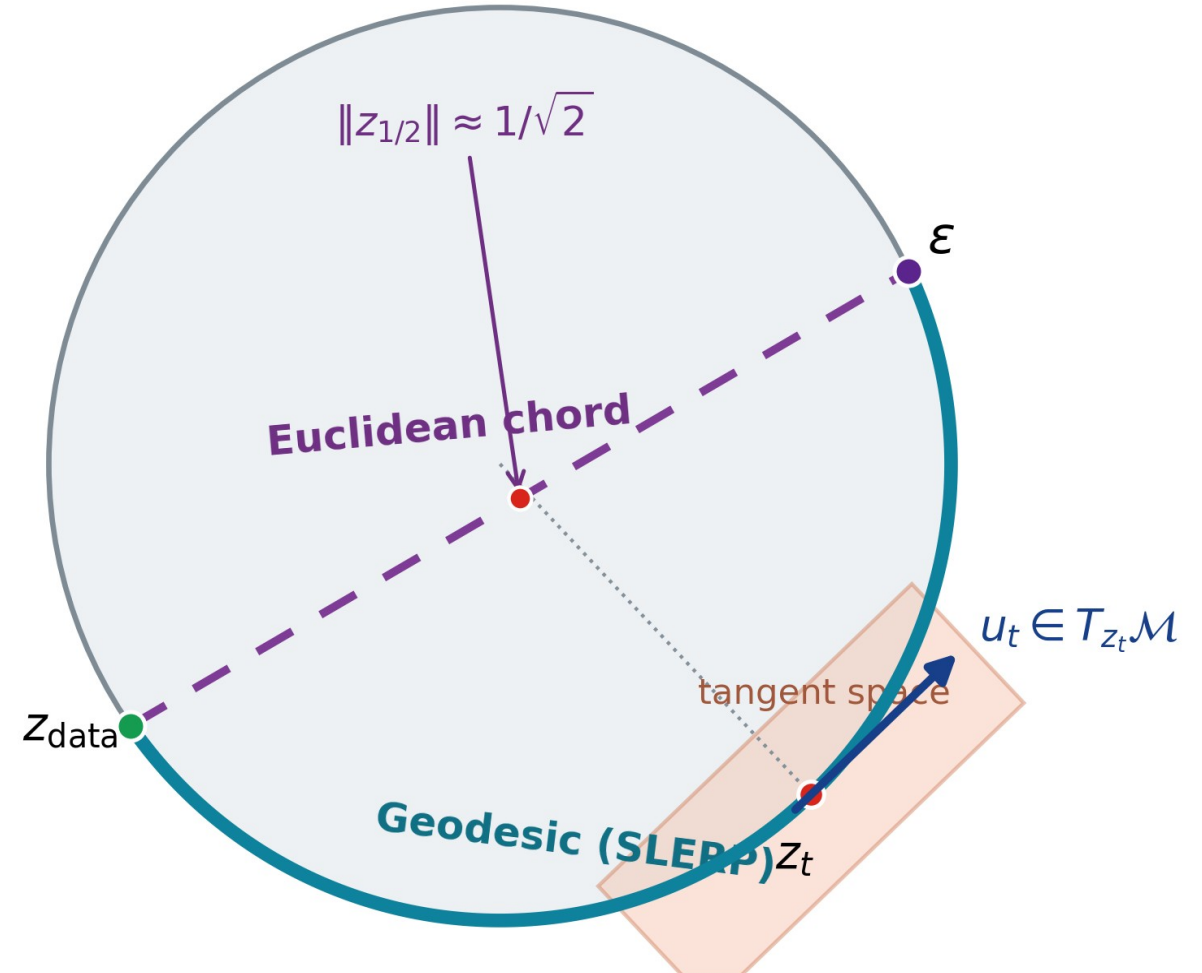
Autoencoder latents need not be Gaussian
Their geometry depends on architecture and training

Normalization can produce a fixed-radius shell
Transformer + LayerNorm is one example

Gaussian noise adds unused radial variation
Linear paths cut through the shell

Research question
For spherical tokens, should transport follow the chord or the geodesic?

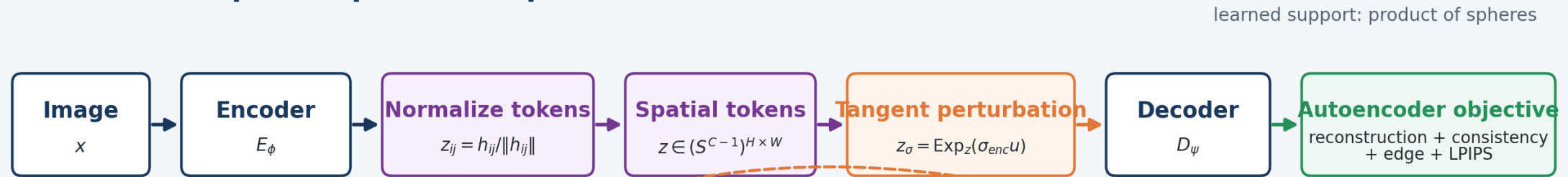
CIFAR-10 • PathMNIST • CelebA-64



SRUL pipeline

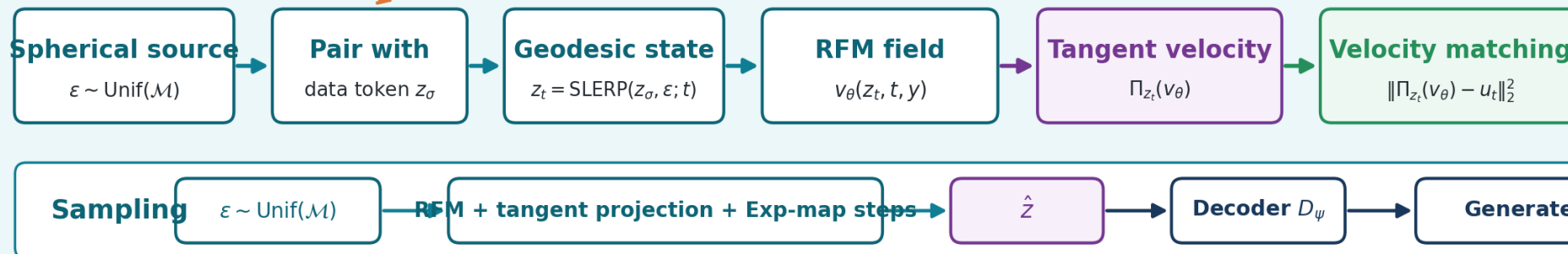
Learn the spherical representation, learn its distribution, then sample on the same manifold

A. Learn the spatial spherical representation



B. Learn and sample the latent distribution with RFM

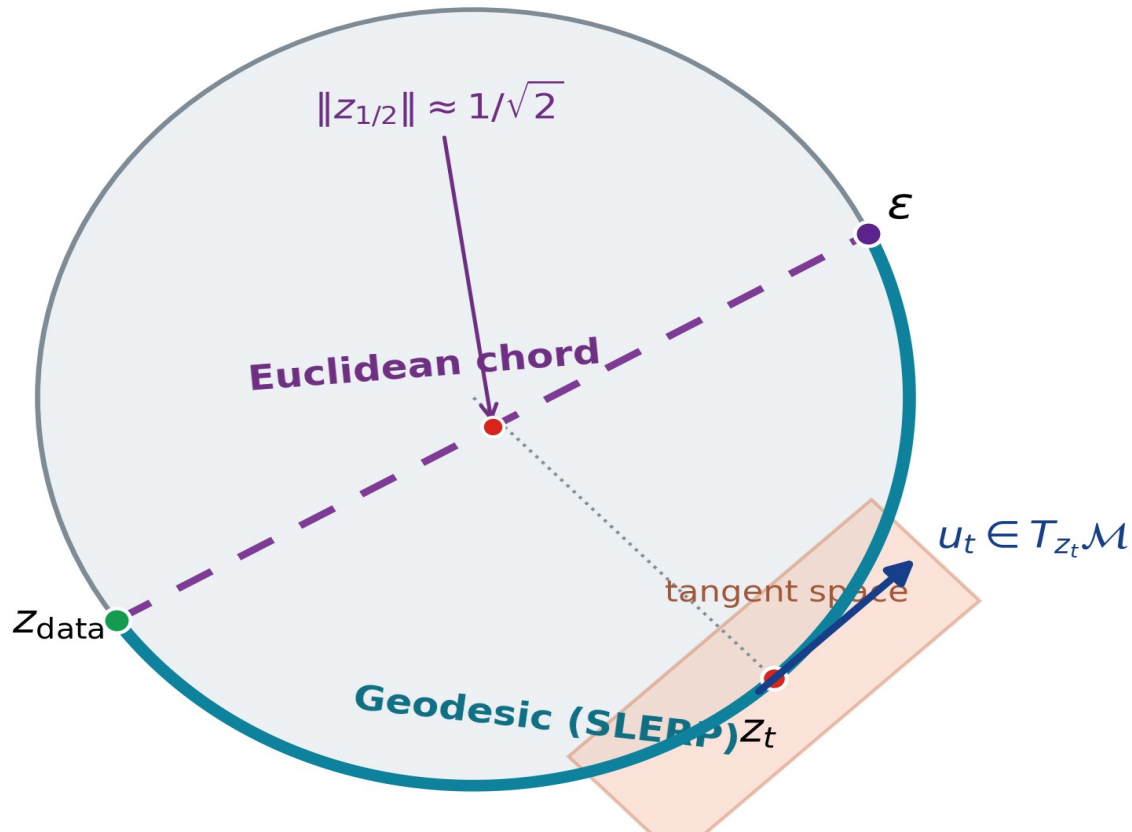
dashed arrow: encoder targets used to train the prior



Core geometry: token normalization + tangent perturbation + SLERP + tangent projection + exponential-map integration.

Controlled test: same spherical autoencoder, different path

Only the transport path changes: Euclidean chord versus SRUL geodesic



Method	FID	KID	Prec.	Recall	Min norm
Chord	67.44	0.0740	0.840	0.085	0.705
SRUL	63.84	0.0667	0.857	0.091	1.000

Why the chord enters the interior

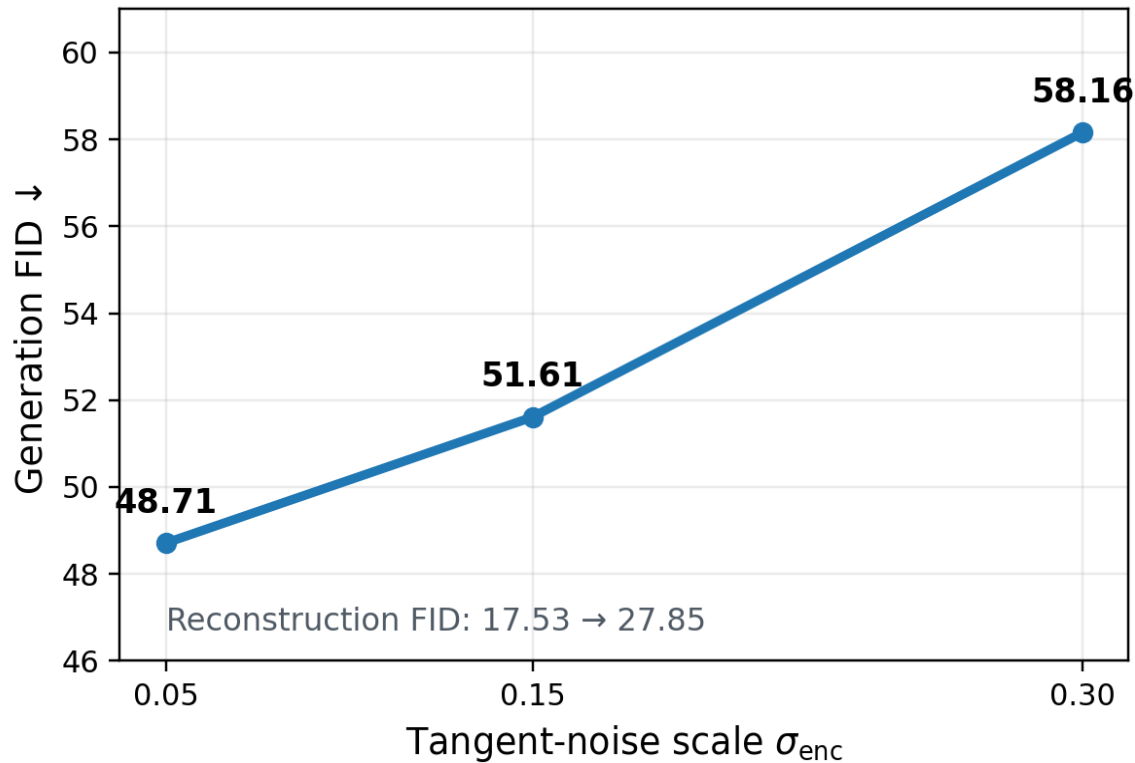
For nearly orthogonal endpoints, the midpoint norm is $1/\sqrt{2} \approx 0.707$. The measured value is 0.705.

With a spherical encoder, geodesic transport improves all four image metrics and keeps the full path on-manifold.

Two independent controls

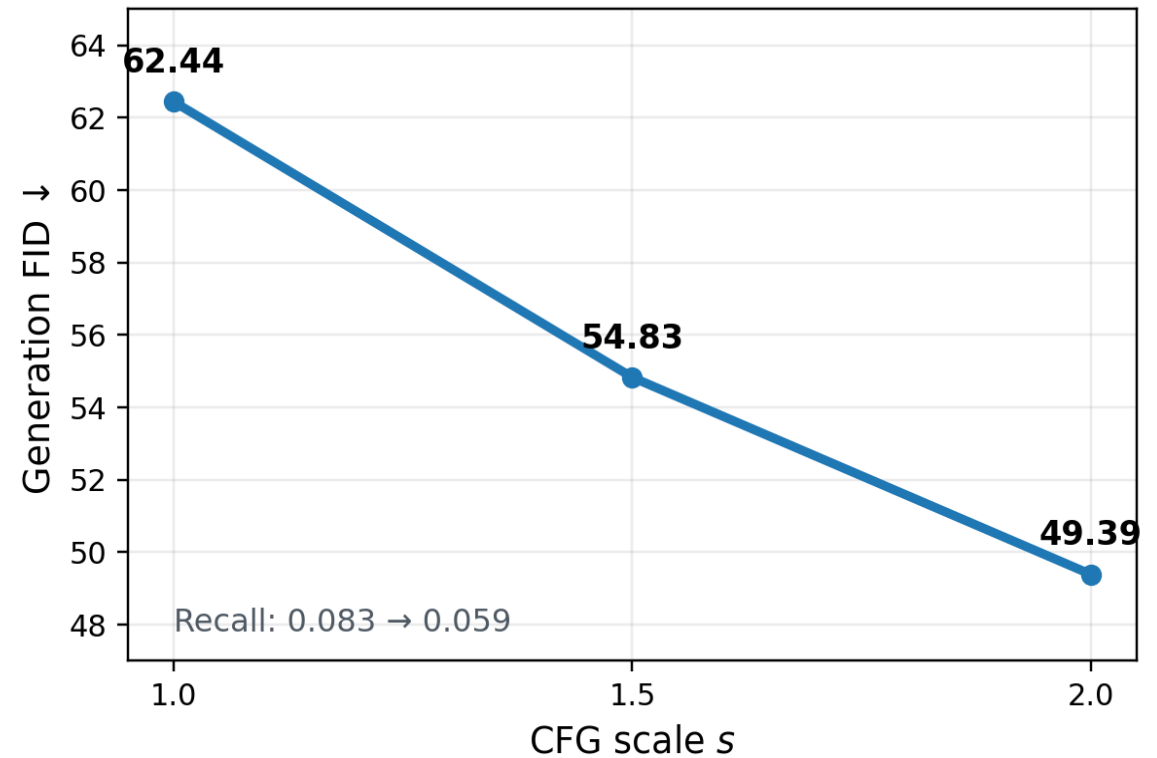
Tangent noise changes the encoded representation; classifier-free guidance changes conditional sampling

Noise ablation



Tangent-noise sweep: $\sigma_{\text{enc}} = 0.05$ gives the best tested reconstruction and generation quality.

Guidance ablation



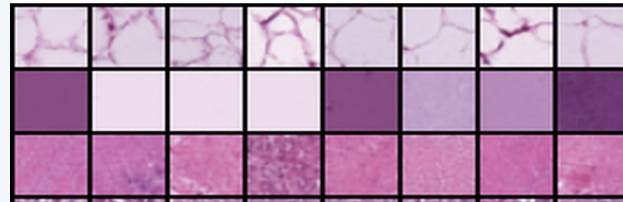
CFG sweep: stronger guidance improves FID and precision, while recall decreases.

Results across three evaluation settings

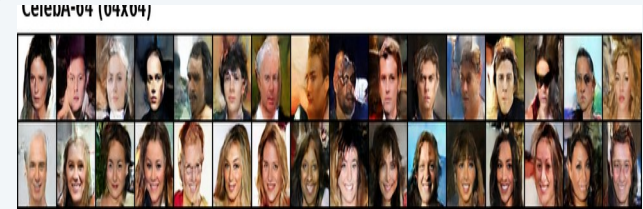
Natural objects, histopathology textures, and structured faces



CIFAR-10



PathMNIST



CelebA-64

Dataset	Evaluation setting	Grid	Recon. FID	Gen. FID	Prec.	Recall
CIFAR-10	Geometry + sampling ablations	4×4	17.53	48.71	0.861	0.090
PathMNIST	Histopathology transfer	4×4	7.13	29.36	0.829	0.309
CelebA-64	64×64 face generation	8×8	9.08	41.22	0.657	0.198

CIFAR-10 provides the controlled ablations; PathMNIST tests texture transfer; CelebA-64 tests resolution and token-grid scaling.